

Quantitative Aptitude

Percentage

Level-1

- Q1** A person spends 35% of his income on education, 12% of the rest on rent, 30 % of the remaining on food and on cloths, if he is left with Rs.18018, find his income.
 (A) Rs 56000 (B) Rs 48000
 (C) Rs 45000 (D) Rs 64000
 (E) Rs 36000
- Q2** If $A^2 = (25\% \text{ of } 324 + 25 \times 4 - 100)$ then find the value of B where $B = 11.11\% \text{ of } 24A$ (Consider only the positive value of A).
 (A) 24 (B) 26
 (C) 28 (D) 32
 (E) 42
- Q3** Mr. Khan invests 12.5% of his monthly salary, i.e., Rs. 14250 in Mutual Funds. Later he invests 12% of his monthly salary on NPS. He invests another 5.5% of his salary on some LIC Policies as well. What is the total monthly amount invested by Mr. Khan?
 (A) 31200 (B) 34000
 (C) 33200 (D) 34100
 (E) 34200
- Q4** If 50% of $(x - y) = 30\% \text{ of } (x + y)$, then what percent of x is y?
 (A) 20% (B) 15%
 (C) 30% (D) 25%
 (E) None of these
- Q5** If 40% of A = 60% of B and B = x% of A, then the value of x is :
 (A) 50 (B) 33.33
 (C) 66.66 (D) 20
 (E) None of these
- Q6** Pooja spends 10% of her monthly income on rent, 20% of the remaining on food, and 50% of the remaining on traveling. If the difference between the amount she spends on traveling and that on rent is Rs.52000, then find the monthly income (in Rs.) of Pooja.
 (A) RS. 70000 (B) RS. 150000
 (C) RS.2000 (D) RS. 5000
 (E) RS. 200000
- Q7** In a Library 60% of the books are in Urdu, 60% of the remaining books are in Hindi. Rest of the books are in English, if there are 9600 books in English, then the total number of books in Urdu is ?
 (A) 6000 (B) 12000
 (C) 36000 (D) 18000
 (E) 10000
- Q8** $\frac{2}{3}$ rd of first number is equal to the second number. If the second number is equal to 24 % of 200 , what is sum of the first and second numbers?
 (A) 125 (B) 60
 (C) 240 (D) 120
 (E) 280
- Q9** The current population of the village is 40000. And the population of this village has increased by 10% per year. Then, find the population of this village after 3 years?
 (A) 53240
 (B) 55000
 (C) 51000
 (D) 51324
 (E) None of the above
- Q10** In an election, two candidates participated, 15% of the votes were declared invalid and the



winner got 60% of the valid votes and won by 3400 votes. Find the number of votes?

- (A) 22000 (B) 15000
(C) 19000 (D) 20000
(E) 18000

Q11 There were 180 students who appeared in the exam, only 35 students were not able to pass the exam. 70 students were able to score first division then how many percent of students stood 2nd or 3rd ?

- (A) 41.67% (B) 60.5%
(C) 33.33% (D) 50.2%
(E) None of these

Q12 Class A has 20% more students than class B. The percentage of girls in class A and B is 60% and 50% respectively. The number of boys in class A is what percent of the number of boys in the class B?

- (A) 88% (B) 96%
(C) 112% (D) 120%
(E) None of these

Q13 The income of X is 40% more than Y's income and the income of Y is 60 % more than Z's income. X's income is how much percent more than Z's income?

- (A) 124% (B) 120%
(C) 122% (D) 130%
(E) None of these

Q14 The monthly expenses of the family on sugar consumption are Rs. 600. If the price of sugar is increased by 66.66%, the family reduces the consumption of sugar by 20 kg, so they make the same sugar expenses after the price hike. Find the initial price per kg of sugar.

- (A) 30 (B) 12
(C) 22 (D) 25
(E) None of these

Q15 In an examination A got 5% marks more than B, B got 10% less than C and C got 20% more than D. If D got 320 marks out of 500, the marks obtained by A were (approx.)?

- (A) 384 (B) 346
(C) 363 (D) 320
(E) None of these

Q16 10% of the voters in an election did not cast their votes. In this election, there were only two candidates. If the winner by obtaining 47% of the total votes defeated his contestant by 1200 votes, then find the total number of voters in the election.

- (A) 20000 (B) 35000
(C) 28000 (D) 30000
(E) None of these

Q17 The sum of two numbers is when added to the 80% of first number and then subtracted from 300% of second number, it gives the sum of $\frac{4}{5}$ of the second number and 120% of the first number. First number is approximately what percent of the second number?

- (A) $85\left(\frac{5}{7}\right)\%$ (B) $87\left(\frac{2}{7}\right)\%$
(C) 80% (D) $84\left(\frac{5}{7}\right)\%$
(E) $83\frac{5}{7}\%$

Q18 A company conducted an exam for placement. The number of girls who appeared for the test is thrice the number of boys. If 40% of the girls and 30% of the boys qualified for the exam, then what percent of students, could not qualify for the exam?

- (A) 62.5% (B) 60%
(C) 56.40% (D) 50%
(E) None of these

Q19 There are 350 students in a school out of which 42 % likes cricket and only 30 students are female who like cricket then what percent of the students are male cricket players?

- (A) 33%
(B) $30\frac{3}{7}\%$
(C) $33\frac{3}{7}\%$
(D) 40%
(E) None of these

Q20



40% of Men are above 30 years of age and 80% of the men are less than or equal to 50 years of age. 20% of all men play basketball. If 30% of the men above the age of 50 play basketball,

what percent of players are less than or equal to 50 years?

- (A) 50%
- (B) 6%
- (C) 70%
- (D) 20%
- (E) None of these



Level-2

- Q1** Fresh blackberries contain $(x + 20)$ percent water while dry blackberries contain $\frac{x}{6}\%$ percent water. If the weight of dry blackberries is 250 kg and weight of fresh blackberries is 1125 kg, then find the quantity of pulp in fresh blackberries.
 (A) **225 kg** (B) 250 kg
 (C) **125 kg** (D) 450 kg
 (E) **None of these**
- Q2** In a class of 1000 students, 65% are girls. 20% of the boys & 40% of the girls failed in the exam. Find the number of students in the school who passed the exam ?
 (A) 650 (B) 350
 (C) 260 (D) 670
 (E) None of these
- Q3** A salesman's commission is 8% on all sales up to Rs.20,000 and 10% on all sales exceeding this . If his total sale in that month was Rs.32000, then find the total amount of commission he made after doing the given sale and the extra sale together.
 (A) 2500 (B) 2200
 (C) 2800 (D) 3020
 (E) None of these
- Q4** Bhagalpur, a poor village, lost 10% of its people due to hungriness, 10% of the remaining due to fever and 10% of the rest were disabled. Thus, the strength was reduced to 7290 active persons. Find the original strength of the village.
 (A) 10000 (B) 12000
 (C) 15000 (D) 18000
 (E) None of these
- Q5** Two friends Rajan and Zahid want to compare their salaries. The combined income of Rajan and Zahid together is Rs. 62500. The income of Rajan and Zahid has increased by 20% and 15% respectively. The new income of Zahid is Rs. 1375 more than the new income of Rajan. What is the new income of Zahid?
 (A) Rs. 37375 (B) Rs. 35625
 (C) Rs. 36500 (D) Rs. 38250
 (E) Rs. 37400
- Q6** In a bank exam, 55% of the students passed in Reasoning and 42% of the students passed in English. If 35% of the students passed in both subjects then find the percentage of students who failed in both subjects.
 (A) 48% (B) 42%
 (C) 38% (D) 36%
 (E) None of these
- Q7** In a College election 20% of the voters on the voters list did not cast their votes, and 120 voters cast their ballot papers blank. There were only two candidates, and the winner was supported by 47% of all the voters in the list and he got 616 votes more than his rival. The number of voters on the list was?
 (A) 5443 (B) 4543
 (C) 3543 (D) 2543
 (E) 2656
- Q8** 11.11% of the soldiers of an army camp are killed in the battle. 16.66% of the remaining soldiers died of corona and 14.28% of the remaining soldiers were disabled. Now only 878802 soldiers were left in the army camp. How many total soldiers were there at the beginning(approx)?
 (A) 1294920 (B) 1394920
 (C) 1494920 (D) 1594920
 (E) 1594920
- Q9** The average of marks scored by Deepak in English, Hindi, and Math is 70 and the average marks scored by Deepak in English and Math are 73. If marks of Deepak in Science are 25% more



than his marks in Hindi, then find the average marks scored by Deepak in these four subjects.

- (A) 76.5 (B) 72.5
(C) 71.5 (D) 75.5
(E) 78.5

Q10 $33\frac{1}{3}\%$ of students in school are male but the number of cricket-playing students is 20% of the total students. If the total females are 750 then what was the total number of students playing cricket in the school?

- (A) 320 (B) 425
(C) 325 (D) 400
(E) None of these

Q11 In a College election 35% voted for Abhi whereas 42% voted for Ravi. The remaining people did not vote for any person. If the difference between those who vote for Ravi in the election and those who are uncertain was 1140, So, how many people participated in college election voting?

- (A) 6000 (B) 3000
(C) 4000 (D) 2000
(E) 5000

Q12 **Directions: Study the following data carefully and answer the questions accordingly.**

In a maths test, Aniket got 414 marks which was 47 less than Baman's marks. The marks of Chand were 48% of the sum of Aniket and Baman's marks together or 52.5% of the total marks. If the marks of Dipen was 32 more than that of Chand's marks. Find how much percent did Dipen get in that examination?

- (A) 48.5% (B) 55.5%
(C) 51.5% (D) 56.5%
(E) None of these

Q13 In a Town, consisting of three areas 1, 2 and 3, the male population of the three areas are in the ratio of 2 : 3 : 4 respectively. In area 1, 55% of the male are working, in area 2, 25% of the male are working and in area 3, 50% of the male are working. Find the percentage (approx.) of non-working male in the Town.

- (A) 32% (B) 57%
(C) 46% (D) 56%
(E) 50%

Q14 In an election only Modi & Gandhi contested, 30% of the voters did not vote and 3200 votes were declared as invalid. The winner, Modi got 9600 votes more than his opponent, thus he secured 51% votes of the total voters on the voter list. Percentage votes of the loser candidate, Gandhi out of the total voters on the voter list is:

- (A) 9% (B) 6%
(C) 12% (D) 3%
(E) None of these

Q15 In an exam. 50% of the students passed in English, and 75% passed in Hindi, While 20% students failed in both the subjects. If 540 students passed in both the subjects, find the total number of students who appeared in the exam?

- (A) 1200 (B) 1400
(C) 1600 (D) 1800
(E) None of these

Q16 A company manufactures 5000 bottles every month. Due to some external factors, 20% of them are found to be defective and sent back to the factory for correction every month. The rest of the bottles are brought to the market for selling. On average the percentage of male buyers of the bottle every month is 30%. Find the number of female buyers of the bottles every month.

- (A) 2400 (B) 2800
(C) 3000 (D) 3600
(E) 3800

Q17 25% of the voters did not cast their vote in an election of Mazdoor Union in Tata company between two candidates. 25% of the votes polled were found invalid. The winning candidate got 54% of the valid votes and won by a majority



of 5240 votes. The number of voters enrolled voters' list was

- (A) 2,17,444 (B) 1,26,445
(C) 1,16,444 (D) 1,56,445
(E) None of these

Q18 Two candidate X and Y contested in an election. 20% of the total votes are declared invalid. Ratio of valid votes got by X and Y is 3:1. Difference between the total votes and valid votes got by X is 400. Find the total number of invalid votes.

- (A) 900 (B) 200
(C) 300 (D) 600
(E) 700

Q19 The total sales done by a salesman is Rs.32000 and the total commission made by him is Rs.3600. If rate of commission up to Rs.20000 is 12% and after Rs,20000 the rate of commission

can either increase or decrease, then find the rate of commission on the sales done over Rs.20000.

- (A) 15% (B) 20%
(C) 10% (D) 12%
(E) None of these

Q20 630 students of a school took an exam. The ratio of boys to girls who took the exam is 3:4. The ratio of number of students passed the exam to those who failed the exam is in the ratio 11:3. Among the girls, the ratio of number of students who passed to those who failed the exam is in the ratio is 11:4. Find the ratio of number of boys to the number of girls who failed?

- (A) 32:15 (B) 16:35
(C) 19:33 (D) 22:39
(E) None of these



Level-3

- Q1** Three machines, known as C_1 , C_2 , and C_3 , contribute to the production of a batch of items. C_1 and C_2 are responsible for 35% and 25% of the production respectively. There is a 2% defect rate for the items produced by C_1 , 1.2% defect rate for C_2 , and a $x\%$ defect rate for C_3 's items. The entire batch of items has a defect rate of $y\%$, where both x and y are integers. ($y < 8$). The task is to determine the total number of possible pairs of x and y .
- (A) 2 Pairs of x and y are possible
 (B) 3 Pairs of x and y are possible
 (C) None of the pairs of x and y are possible
 (D) 1 pairs of x and y are possible
 (E) Both option 2 and 4
- Q2** Salesmen P earns a commission of $Z\%$ on the first Rs. 12000 sale value and $Y\%$ commission on the further same value, while Q earns a commission of $(M - 3)\%$ on the first Rs. 12000 same value and $Y\%$ commission on the further sale value. Total sales given by P and Q are Rs. 18000 and Rs. 20000 respectively, while their earnings are Rs. 240 and 320 respectively. Find $(5M + 4Z)$ is how much % more than that of $10Y$, if the company gives the same commission to each employee on the first Rs. 12000.
- (A) 300% (B) 50%
 (C) 200% (D) 150%
 (E) None of these
- Q3** A has a yearly income of Rs. $24Y$. He spend $X\%$ of his monthly income on house rent, $X+5\%$ of the remaining on food and remaining he saved. If the monthly amount he spent on food and house rent is equal and the difference between the amount he saved and amount spent of food is Rs. 16000, then find then find the value of Y .
- (A) 16,000 (B) 18,000
 (C) 20,000 (D) 22,000
 (E) None of these
- Q4** There are three typist Rahul, Rajat and Ram start typing the book on same time and number of pages in book is different for each typist. Typing rate of Rahul is 25% more than Ram and typing rate of Rajat is 10% less than that of Rahul. Rajat types 1440 pages in A_1 hours which is 8 hours less than time taken by Ram Number of pages type by Rahul is 80 more than that of Ram and A_2 pages more than number of pages type by Rajat . Rate of typing of Rahul is 100 pages/h which is A_3 pages/h more than rate of typing of Ram. The average number of pages typed by all three persons taken together is A_4 . Which of the following given statement(s) can be determined related to the above given data.
- I). Ratio of numerical value of A_2 and A_4
 II). Difference between numerical of A_1 and A_4
 III). Value of A_3
 IV). Value of A_4
- (A) II and III only
 (B) I and II only
 (C) I, II and III only
 (D) III and IV only
 (E) All statements can be determined
- Q5** Ashutosh buys two second hand bikes for Rs 1.5 lakh and Rs 225000 respectively. He spends Rs $(x + 7500)$ on the first bike and Rs ' x ' on the second bike for repairing them. He marks up the first bike by 20% and second bike by 30% and sells the two bikes to Rahul at a discount of 25% each. Now Rahul sells both of them for a total of Rs 540000 at a profit of 20%. Find the amount spent by Ashutosh on repairing both the bikes.
- (A) 102300 (B) 103400
 (C) 100100 (D) 101200
 (E) None of these
- Q6** A salesman imported Laptop worth Rs. 2,50,000, ipad worth Rs. 1,75,000 and iphone Rs. 95,000. He had to pay $x\%$ duty on Laptop, $(x+2)\%$ duty



on ipad and $(x-2)\%$ duty on iphone as special case, where x is the sum of $2a$ and b . If the series $15, a-3, -b+3$, are in AP. Find the total duty in Rs. he had to pay on all items.

- (A) 110,500 (B) 105,500
(C) 1,26,400 (D) 1,00,000
(E) None of these

Q7 From M kg Fresh grapes (contains $Z\%$ water by weight), $\frac{M}{8}$ Kg fresh grapes [which contains $(\frac{Z}{2} - 25)\%$ water by weight] can be formed. Fresh grapes contain $Y\%$ pulp and dry grapes contain $L\%$ pulp.

Which of the following statements is/are true?

- I. $(L - Y) = 70$
II. If 7.5 kg of dry grapes are extracted from M kg of fresh grapes, then the amount of water in fresh grapes is 48 kg.
III. Z is 12.5% more than that of L

- (A) I and II only
(B) II and III only
(C) I and III only
(D) I, II, and III
(E) None of these

Q8 In examination, a student scored 40% marks, after review his score increased by 20% but he still failed by 24 marks. After further review his score was increased by 50% , then he secured 24 marks more than passing marks. If passing marks are Y , then $(Y + 20)$ is how much % of maximum marks?

- (A) 70% (B) 60%
(C) 80% (D) 72%
(E) None of these

Q9 In 2020, the population of town P is twice as that of Q . In 2021, the population of towns P and Q increased by $A\%$ and $B\%$ thus the population of each town became the same. In 2022, the population of town P decreased by 20% , and of Q decreased by 60% . Net increment in population of town P and town Q is $2C\%$ and $(\frac{C}{2} + 15)\%$. Find the value of C .

- (A) 20

- (B) 10
(C) 30
(D) 25
(E) Can't be determined

Q10 A teacher distributed 100 chocolates among three children 'X', 'Y', and 'Z'. The teacher gives $(C - 5)\%$ of the total chocolates to 'X', then he gives $(C + 5)\%$ of the remaining chocolates to 'Y' and finally he gives 25% of the remaining chocolates to 'Z'. The teacher finally left with 54 chocolates. If 'Y' gave 5 chocolates to his younger brother then find the number of chocolates 'Y' left with?

- (A) 8 (B) 13
(C) 9 (D) 17
(E) 20



Answer Key

Level-1

Q1 (C)
Q2 (A)
Q3 (E)
Q4 (D)
Q5 (C)
Q6 (E)
Q7 (C)
Q8 (D)
Q9 (A)
Q10 (D)

Q11 (A)
Q12 (B)
Q13 (A)
Q14 (B)
Q15 (C)
Q16 (D)
Q17 (A)
Q18 (A)
Q19 (C)
Q20 (C)



Level-2

Q1 (A)

Q2 (D)

Q3 (C)

Q4 (A)

Q5 (A)

Q6 (C)

Q7 (C)

Q8 (B)

Q9 (B)

Q10 (E)

Q11 (A)

Q12 (D)

Q13 (B)

Q14 (D)

Q15 (A)

Q16 (B)

Q17 (C)

Q18 (B)

Q19 (C)

Q20 (E)



Level-3

Q1 (B)

Q2 (B)

Q3 (C)

Q4 (E)

Q5 (A)

Q6 (C)

Q7 (C)

Q8 (A)

Q9 (B)

Q10 (B)



[Android App](#)

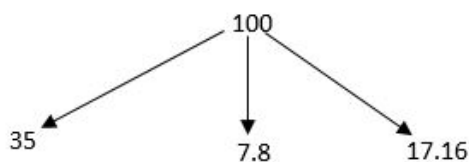
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Hints & Solutions

Level-1

Q1 Text Solution:



Let his income by Rs 100

Expenditure on education = 35

Remaining = 65

Expenditure on rent = 12% of 65 = 7.80

Remaining = 65 - 7.80 = 57.2

Expenditure on food and cloths = 30% of 57.2 = 17.16

Remaining = 57.2 - 17.16 = 40.04

If his saving is Rs 40.04, then his income is Rs 100

When his savings are 18018, then his income = $\frac{18018 \times 100}{40.04} = 45000$

Q2 Text Solution:

Ans. 24

$$A^2 = (25\% \text{ of } 324 + 25 \times 4 - 100)$$

$$A^2 = 0.25 \times 324 + 100 - 100$$

$$A^2 = 81$$

$$A = 9$$

Now,

$$B = 11.11\% \text{ of } 24A$$

$$B = \frac{1}{9} \times 24 \times 9$$

$$B = 24$$

Q3 Text Solution:

Let Mr. Khan's monthly income be x.

Then, 12.5% of x = 14250

$$x = 114000$$

Total monthly amount invested by Mr. Khan =

$$12.5\% + 12\% + 5.5\% = 30\%$$

$$\therefore 30\% \text{ of } 114000 = \text{Rs. } 34200$$

$\therefore$ The answer is Rs.34200.

Q4 Text Solution:

Given;

$$50\% \times (x - y) = 30\% \times (x + y)$$

$$\frac{1}{2} \times (x - y) = \frac{3}{10} \times (x + y)$$

$$5x - 5y = 3x + 3y$$

$$2x = 8y$$

$$\frac{x}{y} = \frac{4}{1}$$

$$\text{Therefore ; } y\% \text{ of } x = \frac{1}{4} \times 100 = 25\%$$

Q5 Text Solution:

$$40\% \text{ of } A = 60\% \text{ of } B$$

$$\frac{40}{100} \text{ of } A = \frac{60}{100} \text{ of } B$$

$$2A = 3B$$

$$\frac{A}{B} = \frac{3}{2}$$

$$\text{Since ; } B = x\% \text{ of } A$$

$$2 = \frac{x}{100} \times 3$$

$$x = \frac{2}{3} \times 100 = 66.66\%$$

Q6 Text Solution:

Let the monthly income of Pooja = 100x

$$\text{So, total amount spends on Rent} = 100x \times \frac{10}{100} = 10x$$

$$\text{Spent of food} = 20\% \text{ of } 90x = 18x$$

$$\text{Total amount spends on travel} = 36x$$

ATQ –

$$36x - 10x = 52000$$

$$x = 2000$$

So, a monthly income of Pooja = Rs.200000

Q7 Text Solution:

Let the number of books in library = x,

Then, Number of Urdu books = 0.6x,

Remaining books = x - 0.6x = 0.4x

$\Rightarrow$ 60% of remaining books is Hindi =

$$0.6 \times 0.4x = 0.24x$$

$$\Rightarrow \text{Number of English books} = X - 0.6X - 0.24x = 0.16x$$

$$\text{Given, } 0.16X = 9600.$$

$$\text{So, } X = 60000$$

$$\text{Hence, No. of Urdu books} = 0.6 \times 60000 = 36000$$

Q8 Text Solution:

$$\text{Second number} = \frac{200 \times 24}{100} = 48$$

$$\text{First number} = 48 \times \frac{3}{2} = 72$$

$$\text{Required sum} = 48 + 72 = 120$$

$\therefore$ The answer is 120.



Q9 Text Solution:

Let be the current population of the village P

Population after 3 years = $p(1 + \frac{r}{100})^t$

$$= 40000(1 + \frac{10}{100})^3$$

$$= 40000 \times \frac{11}{10} \times \frac{11}{10} \times \frac{11}{10}$$

$$= 53240$$

∴ The answer is 53240.

Q10 Text Solution:

Given:

15% of the votes were declared invalid

the winner gets 60% of the valid votes

Total no. of votes = Valid votes + Invalid votes

Let the total number of votes be $100x$

$$\text{Valid votes} = 100 - 15x = 85x$$

Now the winner gets 60% of valid votes and the loser gets 40% votes

Therefore, difference between winner and loser votes = 60% - 40% of $85x$

$$20\% \text{ of } 85x$$

$$17x$$

$$\text{And } 17x = 3400$$

$$x = 200$$

$$100x = 20000$$

The number of total votes is 20000

Q11 Text Solution:

There are total of 180 students.

Only 35 students did not pass the exam.

$$\text{So, no. of students passed the exam} = 180 - 35 = 145$$

And 70 students stood first, so the no. of students stood

$$2\text{nd or } 3\text{rd} = 145 - 70 = 75$$

$$\% \text{ of students stood } 2\text{nd or } 3\text{rd} = \frac{75}{180} \times 100 = 41.67\%$$

Q12 Text Solution:

Let the number of students in class B be 100.

The number of students in class A = 120% of B = 120.

The number of girls in class A = 60% of 120 = 72

The number of girls in class B = 50% of 100 = 50

The number of boys in class A = 120 - 72 = 48

The number of boys in class B = 100 - 50 = 50

$$\text{Required percentage} = \frac{48}{50} \times 100 = 96\%$$

Q13 Text Solution:

Let Z's income = 100

Then, Y's income

$$= 160\% \times Z = 100 + 60 = 160$$

So, X =

$$(100 + 40)\% \text{ of } Y = 160 + 64 = 224$$

$$\text{So, X more than Z} = 224 - 100 = 124\%$$

Q14 Text Solution:

Let the initial price of sugar be 100 units and the new price be 166.66 units

Now given that, If the price of sugar is increased by 66.66%, the family reduces sugar consumption by 20 kg.

$$\therefore 166.66 \text{ units} - 100 \text{ units} = 20 \text{ kg}$$

$$\Rightarrow 66.66 \text{ units} = 20 \text{ kg}$$

$$\Rightarrow 1 \text{ unit} = \frac{20}{66.66} = 0.30 \text{ kg}$$

$$\Rightarrow 100 \text{ units} = \frac{20}{66.66} \times 100 = 30 \text{ kg (New consumption)}$$

$$\therefore \text{Initial Consumption} = 30\text{kg} + 20 \text{ kg} = 50 \text{ kg}$$

$$\therefore \text{The initial price per kg of sugar} = \frac{600}{50} = 12 \text{ Rs.}$$

Q15 Text Solution:

Maximum marks = 500

Total marks of D = 320

$$\text{Marks of C} = (100+20)\% \text{ of D} = 120\% \text{ of } 320 = \frac{120}{100} \times 320 = 384$$

$$\text{Marks of B} = (100 - 10)\% \text{ of C} = 90\% \text{ of } 384 = \frac{90}{100}$$

$$\times 384 = 345.6 \approx 346$$

$$\text{Marks of A} = (100+5)\% \text{ of B} = 105\% \text{ of } 345.6 = \frac{105}{100} \times 345.6 = 362.88 \approx 363$$

Q16 Text Solution:

Let V be the total number of voters in the election, N be the number of voters who did not cast their votes, x be the number of votes received by the losing candidate, y be the number of votes received by the winning candidate.

We are given that:

$$N = 0.1V \text{ (10\% of the voters in the election did not cast their votes)}$$



$y = 0.47V$ (the winner obtained 47% of the total votes)

$y - x = 1200$ (the winner defeated his contestant by 1200 votes)

total number of votes cast

$$\Rightarrow x + y + N = V$$

Substituting the values we know, we get:

$$x + 0.47V + 0.1V = V$$

Simplifying this equation, we get:

$$x = 0.43V$$

$$\text{Put } x = 0.43V \text{ in } y - x = 1200$$

$$y - 0.43V = 1200$$

$$0.47V - 0.43V = 1200$$

$$0.04V = 1200$$

$$V = 30000$$

Therefore, the total number of voters in the election is 30,000.

Q17 Text Solution:

Let first number = a Second number = b

The sum of two numbers is when added to the 80% of first number and then subtracted from 300% of second number, it gives the sum of second number and 120% of first number, i.e.,

$$300\% \text{ of } b - \{(a + b) + 80\% \text{ of } a\} = \left(\frac{4b}{5}\right)$$

+ a 120% of a

$$6b = 7a \quad \left(\frac{a}{b}\right) = \left(\frac{6}{7}\right)$$

$$\text{Percentage} = \left(\frac{a}{b}\right) \times 100 = \left(\frac{6}{7}\right) \times 100$$

$$= 85\left(\frac{5}{7}\right)\%$$

Q18 Text Solution:

Let the number of boys = x and girls = 3x

Number of girls who could not qualify for the exam = 60% of 3x = 1.8x

Number of boys who could not qualify for the exam = 70% of x = 0.7x

Total number of students who could not qualify for the exam = 1.8x + 0.7x = 2.5x

Total number of students = 3x + x = 4x

$$\text{Required \%} = \frac{2.5x}{4x} \times 100 = 62.5\%$$

Q19 Text Solution:

Total students = 350

No. of students like cricket = 42 % of 350 = 147

No. of female cricket player = 30

So, male cricketers = 147 - 30 = 117

$$\text{Required \%} = \frac{117}{350} \times 100 = \frac{234}{7} = 33\frac{3}{7}\%$$

Q20 Text Solution:

Let, total men = 100

Men less than or equal to 50 years = 80 and men above 50 years = 20

No. of men plays basketball = 20

30% of the men above 50 plays basketball = 6

So, remaining 14 men who plays basketball are less than or equal to 50 years

$$\text{So, Required \%} = \frac{14}{20} \times 100 = 70\%$$



Level-2

Q1 Text Solution:

We know that , weight of pulp in fresh blackberries = weight of pulp in dry blackberries

$$1125 \times \frac{(100 - (x + 20))}{100} = 250 \times \frac{100 - \frac{x}{6}}{100}$$

$$1125 \times (100 - x - 20) = \frac{250(600 - x)}{6}$$

$$3375(80 - x) = 75000 - 125x$$

$$3375x - 125x = 3375 \times 80 - 75000$$

$$3250x = 270000 - 75000 = 19500$$

$$x = 60$$

Quantity of pulp in fresh blackberries =

$$1125 \times \frac{100 - (60 + 20)}{100} = 1125 \times \frac{20}{100} = 225 \text{ kg}$$

Q2 Text Solution:

$$\text{Total students} = 1000 = 100\%$$

$$\text{Number of girls} = 65\% \text{ of } 1000 = 650$$

$$\text{Number of boys} = 35\% \text{ of } 1000 = 350$$

$$\text{Number of Boys failed in exam} =$$

$$350 \times \frac{20}{100} = 70$$

$$\text{Number of girls failed in exam} = 650 \times \frac{40}{100} = 260$$

$$\text{Total Number of students fail} = 260 + 70 = 330$$

$$\text{Total Number of students passed} = 1000 - 330 = 670$$

Q3 Text Solution:

$$\text{Commission} \quad \text{on}$$

$$20000 = 8\% \text{ of } 20000 = 1600$$

$$\text{Remaining sales} = 32000 - 20000 = 12000$$

$$\text{Commission} \quad \text{on}$$

$$12000 = 10\% \text{ of } 12000 = 1200$$

$$\text{Total commission} = 1200 + 1600 = 2800$$

Q4 Text Solution:

Let there be 100 persons in the beginning.

$$\text{After losing } 10\% \text{ of } 100 \text{ due to hunger} = 100 - 10 = 90$$

$$\text{After losing } 10\% \text{ of remaining due to fever} = (100 - 10)\% \text{ of } 90 = 81$$

$$\text{After losing } 10\% \text{ of rest due to disability}$$

$$= (100 - 10)\% \text{ of } 81 = 72.9$$

$$\text{Now, } (72.9 : 100 :: 7290 : x)$$

$$x = \frac{100 \times 7290}{72.9} = 10000$$

Hence, option (A) is correct

Q5 Text Solution:

Answer: A

Solution:-

Let the income of Rajan and Zahid be R and Z,

The monthly income of Rajan and Zahid = 62500

$$R + Z = 62500$$

$$\text{Rajan's income} = x; \text{Zahid's income} = 62500 - x$$

$$\text{New income of Z} = \text{New income of R} + 1375$$

$$Z's \text{ new income} = (62500 - x) \times \frac{115}{100}$$

$$R's \text{ new income} = x \times \frac{120}{100}$$

$$(62500 - x) \times \left(\frac{115}{100}\right) = x \times \left(\frac{120}{100}\right) + 1375$$

$$\frac{(7187500 - 115x)}{100} = \left(\frac{120x}{100}\right) + 1375$$

$$\frac{(7187500 - 115x)}{100} = \frac{(120x + 137500)}{100}$$

$$(7187500 - 115x) = (120x + 137500)$$

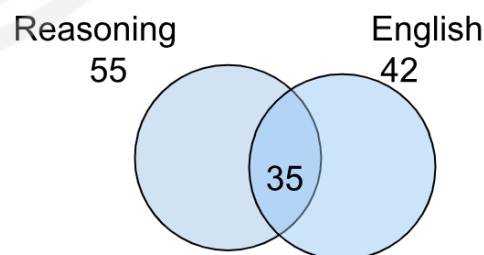
$$7187500 - 137500 = 115x + 120x$$

$$7050000 = 235x$$

$$\text{Rajan's income } X = \frac{7050000}{235} = 30000$$

$$\text{Zahid's income} = 62500 - x = 32500$$

$$\text{New Income of Zahid} = 32500 \times \frac{115}{100} = \text{Rs. } 37375$$

Q6 Text Solution:

$$\text{Total passed students only in Reasoning} = 55 - 35 = 20\%$$

$$\text{Total passed only in English} = 42 - 35 = 7\%$$

$$\text{Let total number of students} = 100\%$$

$$\text{Therefore, required percentage of students who failed in both subjects} = 100 - (20 + 7 + 35) = 38\%$$

Q7 Text Solution:

Let the total no. of voters = x

People who voted for the winner are = 47% of total voters

$$= 0.47 \times x = 0.47x$$

People who voted for loser = $0.47x - 616$

People casted blank votes = 120

And, People who did not vote = 20% of the total voter

$$= 0.2 \times x = 0.2x$$

Adding all the votes and non votes and equating to total voters list,

$$\text{So, } 0.47x + 0.47x - 616 + 120 + 0.2x = x$$

$$x \approx 3543$$

$\therefore$ Hence the answer is 3543.

Q8 Text Solution:

Let, Initial number of soldiers = x

Then, soldiers left after battle = $(100 - 11.11)\%$ of x
 $= 88.89\%$ of x

$$= 0.8889x$$

Again, Number of soldiers left after corona

$$= (100 - 16.66)\% \text{ of } 0.8889x = 0.74x$$

Also, Number of soldiers left after disabled soldiers are removed

$$= (100 - 14.28)\% \text{ of } 0.74x = 0.63x$$

Now, Given $0.63x = 878802$

$$\text{Hence, } x = 1394920$$

Q9 Text Solution:

Marks of Deepak in Hindi = $(70 \times 3) - (73 \times 2) = 64$

$$\text{Marks of Deepak in Science} = 64 \times \frac{125}{100} = 80$$

$$\text{Required average} = \frac{64 + 80 + 73 \times 2}{4} = 72.5$$

Q10 Text Solution:

$33\frac{1}{3}\%$ of students in school are male

So, number of female = $66\frac{2}{3}\%$ can be written as $\frac{2}{3}$.

Given The number of females is equal to 750.

We can assume that if the total student is $3x$ then the number of females will be $2x$ and so

$$2x = 750$$

$$\text{So, } x = 375$$

$$\text{Thus total number of student} = 3x = 3 \times 375$$

And the number of students playing cricket is 20% i.e one-fifth of total students.

$$\text{So, required number} = \frac{1}{5} \times 3 \times 375 = 225$$

Hence, the answer is 225.

Q11 Text Solution:

Let, Total no. of people who voted in the election = x

$$\text{Percent of students who didn't vote} = 100 - (35 + 42) = 23\%$$

And, the difference between who voted for ravi and those who were uncertain = 42% of x - 23% of $x = 1140$

$$19\% \text{ of } x = 1140$$

$$\text{So, } x = 6000$$

Q12 Text Solution:

From the given information,

$$\text{Baman's marks} = 414 + 47 = 461$$

$$\text{The sum of Aniket's and Baman's marks} = 414 + 461 = 875$$

$$\text{Chand Marks} = 48\% \text{ of } 875 = 420 = 52.5\% \text{ of the total marks} = 52.5\% \text{ of } a \text{ (let the total marks is } a \text{)}$$

$$\text{By solving, } a = \text{total marks} = 800$$

$$\text{Dipen's marks} = 420 + 32 = 452$$

$$\text{Required percentage} = \frac{452 \times 100}{800} = 56.5\%$$

Q13 Text Solution:

Let, Number of males in area 1, 2 & 3 are $2x$, $3x$, & $4x$ respectively

$$\text{So, total number of males} = 2x + 3x + 4x = 9x$$

$$\text{Number of working men in area 1} = 55\% \text{ of } 2x = \frac{55}{100} \times 2x = 1.1x$$

$$\text{Number of working men in area 2} = 25\% \text{ of } 3x = \frac{25}{100} \times 3x = 0.75x$$

$$\text{Number of working men in area 3} = 50\% \text{ of } 4x = \frac{50}{100} \times 4x = 2x$$

$$\text{Total number of working men} = 1.1x + 0.75x + 2x = 3.85x$$

$$\text{So, total number of non-working men} = 9x - 3.85x = 5.15x$$

$$\text{Hence, percentage of non-working males} = \frac{5.15x}{9x} \times 100 = 57.22\% \approx 57\%$$



Q14 Text Solution:

Let, total voters = x

Given, Number of people who didn't vote = 30% of $x = 0.3x$

Number of invalid votes = 3200

Modi got 51% votes of total voter list = 51% of $x = 0.51x$

No. of votes Gandhi got = 9600 less than Modi = $0.51x - 9600$

So, total votes = $x = 0.3x + 3200 + 0.51x + 0.51x - 9600$

$$x = 20000$$

So, No. of votes of modi = $0.51x = 10200$

And, No. of people who didn't vote = $0.3x = 6000$

So, No. of votes of gandhi = $10200 - 9600 = 600$

Hence, required% = $\frac{600}{20000} \times 100 = 3\%$

Q15 Text Solution:

Let total No. of students = x

Number. of students Passed in English = 50% of $x = 0.5x$

Number of students passed in Hindi = 75% of $x = 0.75x$

Number of students failed in both the subjects = 20% of $x = 0.2x$

Number of students passed in at least one subject = $x - 0.2x = 0.8x$

And, No. of students passed in both the subjects = 540

Also, No. of students passed in both the subjects = $(50\% + 75\% - 80\%)$ of $x = 45\%$ of x

So, 45% of $x = 540$.

$$\text{Hence, } x = \frac{540 \times 100}{45} = 1200$$

Q16 Text Solution:

Number of bottles brought to the market for selling = $5000 - 5000 \times \frac{20}{100} = 4000$

Percentage of male buyers of the bottle = 30%

Therefore, the percentage of the female buyer of the bottle = 70%

Number of female buyers of the bottles every month $\frac{70}{100} \times 4000 = 2800$

Q17 Text Solution:

Let the total number of voters be x . Then, votes polled

= 75% of x .

Valid votes = 75% of (75% of x).

$\therefore$ 54% of [75% of (75% of x)] – 46% of [75% of (75% of x)] = 5240

$$\Leftrightarrow 8\% \text{ of } [75\% \text{ of } (75\% \text{ of } x)] = 5240$$

$$\Leftrightarrow \frac{8}{100} \times \frac{75}{100} \times \frac{75}{100} \times x = 5240$$

$$\Leftrightarrow x = \left(\frac{5240 \times 100 \times 100 \times 100}{8 \times 75 \times 75} \right) = 1,16,444$$

Q18 Text Solution:

Let total votes be 100t.

Given, 20% of the total votes are declared invalid.

So, total valid votes =

$$\left(\frac{80}{100} \right) \times 100t = 80t$$

$$\text{Valid votes got by X} = \frac{3}{4} \times 80t = 60t$$

$$\text{Valid votes got by Y} = \frac{1}{4} \times 80t = 20t$$

According to the question,

$$100t - 60t = 400$$

$$40t = 400$$

$$t = 10$$

$$\text{Required value} = 20t = 20 \times 10 = 200$$

Q19 Text Solution:

Ans. 10%

Let's assume the rate of commission on the sales above Rs. 20,000 is 'x' percent.

We can divide the total sales into two parts: sales up to Rs. 20,000 and sales above Rs. 20,000.

For, Sales up to Rs. 20,000:

Commission

$$= 12\% \text{ of Rs. } 20,000 = \text{Rs. } 2,400$$

For, Sales above Rs. 20,000:



Commission

$$= x\% \text{ of } (Rs. 32,000 - Rs. 20,000)$$

$$= x\% \text{ of } Rs. 12,000$$

$$= Rs. \frac{x}{100} \times 12,000 = Rs. 120x$$

Total

Commission

$$= Rs. 2,400 + Rs. 120x = Rs. 3,600$$

Simplifying the above equation, we get:

$$Rs. 120x = Rs. 3,600 - Rs. 2,400$$

$$Rs. 120x = Rs. 1,200$$

$$x = 10$$

Therefore, the rate of commission on the sales done above Rs. 20,000 is 10%.

Q20 Text Solution:

No. of students = 630

$$\text{No. of boys} = 630 \times \frac{3}{7} = 270$$

No. of girls = 360

$$\text{Total passing students} = 630 \times \frac{11}{14} = 495$$

$$\text{Total failing students} = 630 - 495 = 135$$

$$\text{Total girls passing} = \frac{11}{15} \times 360 = 264$$

$$\text{So no. of boys passing} = 495 - 264 = 231$$

$$\text{No. of girls failing} = 360 - 264 = 96$$

$$\text{And boys failing} = 270 - 231 = 39$$

$$\text{Required ratio} = 39:96 = 13:32$$

Hence, option e.



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Level-3

Q1 Text Solution:

Let the number of total goods be 1000a

C_1 produces 35% goods = 350a

C_2 produces 25% goods = 250a

C_3 produces 40% goods = 400a

Defective products by $C_1 = 7a$

Defective products by $C_2 = 3a$

Defective products by $C_3 = 4a$

It is given overall defective products are y%

Therefore $7a + 3a + 4a = 10ay$

$\Rightarrow 10 + 4x = 10y$

It is given x, and y are natural numbers such that $y < 8$

$\Rightarrow 4x = 10y - 10$

RHS is always a number multiple of 10

The value of x should be such that LHS is multiple of 10.

Therefore, $x = 5, 10, 15, \dots$

If $x = 5, y = 3$

If $x = 10, y = 5$

If $x = 15, y = 7$

For the remaining values of x, y is greater than 8.

Therefore, only 3 pairs of x and y are possible.

Q2 Text Solution:

According to the question,

$Y\% \times (20000 - 12000) - Y\% \times (18000 - 12000)$

$= 920 - 840$

$20 \times Y = 80$

Value of $Y = 4$

Also,

$Z = M - 3 \dots \dots \dots (1)$

Now,

$Z\% \times 12000 + 4\% \times 6000 = 840$

$120Z = 600$

Value of $Z = 5$

So, value of $M = 5 + 3 = 8$

Value of $(5M + 4Z) = 5 \times 8 + 4 \times 5 = 60$

Value of $10Y = 10 \times 4 = 40$

Required % change = $\frac{(60-40)}{40} \times 100 = 50\%$

Hence answer is option B.

Q3 Text Solution:

The monthly income of A = $\frac{24Y}{12} = 2Y$

Amount spend on house rent = $2Y \times \frac{X}{100}$

Amount spend on food =

$2Y \times \left(\frac{100-X}{100} \right) \times \left(\frac{X+5}{100} \right)$

According to question,

$2Y \times \frac{X}{100} = 2Y \times \left(\frac{100-X}{100} \right) \times \left(\frac{X+5}{100} \right)$

$100X = (100 - X)(X + 5)$

$x^2 + 5x - 500 = 0$

$X = -25, 20$

We cannot consider the negative value

So, $X = 20$

Amount he saved

$= 2Y \times \left(\frac{100-X}{100} \right) \times \left(\frac{100-X-5}{100} \right)$

$= 2Y \times \left(\frac{80}{100} \right) \times \left(\frac{75}{100} \right)$

$2Y \times \left(\frac{80}{100} \right) \times \left(\frac{75}{100} \right) - 2Y \times \left(\frac{80}{100} \right) \times \left(\frac{25}{100} \right)$

$= 16000$

$2Y \times \left(\frac{80}{100} \right) \times \left(\frac{75}{100} - \frac{25}{100} \right) = 16000$

$Y = 20,000$

Q4 Text Solution:

Rate of typing of Rahul = 100 pages/h

Rate of typing of Ram = $\frac{100}{125} \times 100 = 80$ pages/h

Rate of typing of Rajat = 90% of 100 = 90 pages/h

Value of $A_3 = 100 - 80 = 20$ pages/h

Time taken by Rajat to type 1440 pages = $\frac{1440}{90} = 16$ hours

Value of $A_1 = 16$ hours

Time taken by Ram = $16 + 8 = 24$ hours

Number of pages typed by Ram = $24 \times 80 = 1920$ pages

So, number of pages type by Rahul = $1920 + 80 = 2000$ pages

Value of $A_2 = 2000 - 1440 = 560$ pages

Value of $A_4 = \frac{(2000+1440+1920)}{3} = 1786.66$ pages

So All statements can be determined

Q5 Text Solution:

Cost price of first bike

$$\text{Rs. } 1.5 \text{ Lakh} + \text{Rs. } (x+7500) = \text{Rs. } (x+157500)$$

Cost price of second bike

$$= \text{Rs } 225000 + \text{Rs } x$$

$$= \text{Rs } (225000 + x)$$

Total cost price for Ashutosh =

$$(x + 157500) + (225000 + x) = 2x$$

$$+ 382500$$

Ashutosh marked up the first bike by 20%, so

MP of first bike

$$= 1.2(x + 157500) = 1.2x + 189000$$

Similarly,

MP of second bike

$$= 1.3(225000 + x) = 292500 + 1.3x$$

Since Rahul bought both bikes at a 25% discount, the total selling price is:

Total selling price = $0.75(\text{MP of first bike} + \text{MP of second bike})$

$$= 0.75(1.2x + 189000 + 292500$$

$$+ 1.3x)$$

$$= 1.875x + 361125$$

Rahul sells both bikes at a profit of 20%, so his total revenue is 1.2 times the total selling price:

Total revenue for Rahul = $1.2(\text{Total selling price})$

$$= 1.2(1.875x + 361125)$$

$$= 2.25x + 433350$$

We know that the total revenue for Rahul is Rs 540000, so:

$$2.25x + 433350 = 540000$$

$$x = 47400$$

Therefore, the amount spent by Ashutosh on repairing both bikes is:

Amount spent

$$= x + (x + 7500) = 2x + 7500$$

$$= \text{Rs } 102300$$

Q6 Text Solution:

The AP series is 15, $a-3$, $-b+3$

Common difference,

$$a - 3 - 15 = -b + 3 - a + 3$$

$$2a - 18 = -b + 6$$

$$2a + b = 24$$

The value of x is 24.

The Duty on Laptop

$$2,50,000 \times \frac{24}{100} = 60,000$$

Duty on iPad

$$1,75,000 \times \frac{26}{100} = 45,500$$

Duty on iPhone

$$95,000 \times \frac{22}{100} = 20,900$$

The total duty paid by salesman.

$$60,000 + 45,500 + 20,900 = 126,400$$

Q7 Text Solution:

The amount of pulp in fresh grapes and dry grapes must be the same.

$$M \times (100 - Z) \% = \frac{M}{8} \times (100 - \frac{Z}{2} + 25) \%$$

$$800 - 8Z = 125 - \frac{Z}{2}$$

$$675 = \frac{15Z}{2}$$

So, the value of $Z = 90$

$$\text{Value of } Y = 100 - 90 = 10$$

$$\% \text{ of water in dry grapes} = \frac{90}{2} - 25 = 20\%$$

$$\text{Value of } L = 100 - 20 = 80$$

$$I.(L - Y) = 70$$

$$\text{Required value} = 80 - 10 = 70$$

This statement is true.

II. If 7.5 kg of dry grapes are extracted from M kg of fresh grapes, then the amount of water in fresh grapes is 48 kg.

$$\text{In 7.5 kg dry grapes, the amount of pulp} = 80\% \times 7.5 = 6 \text{ kg}$$

The pulp in fresh grapes = 6 kg

$$\text{Amount of water in fresh grapes} = \frac{6}{10} \times 90 = 54 \text{ units}$$

This statement is false.

III. Z is 12.5% more than that of L

$$\text{Required \% change} = \frac{(90-80)}{80} \times 100 = 12.5\%$$

This statement is true.

Hence answer is option C

Q8 Text Solution:

Initial score of student = 40%

$$\text{Score after first review} = 40\% \times 120\% = 48\%$$

$$\text{Score after second review} = 48\% \times 1.5 = 72\%$$

Now,

$$= 48\% + 24 = 72\% - 24$$

$$= 24\% = 48$$



$$\text{So, maximum marks} = \frac{48}{24} \times 100 = 200$$

$$\text{Value of Y} = 48\% \times 200 + 24 = 120$$

$$\begin{aligned} \text{Required percentage} &= \frac{120+20}{200} \times 100 \\ &= 70\% \end{aligned}$$

Q9 Text Solution:

In 2021 population of both is the same.

Let the population of town Q in 2020 = a

So, the population of town P in 2020 = $2 \times a = 2a$

Now,

$$\frac{2a \times (100 + 2C)\%}{a \times \left(100 + \frac{C}{2} + 15\right)\%} = \frac{(100 - 20)\%}{(100 - 60)\%}$$

$$2C = \frac{C}{2} + 15$$

$$\frac{3C}{2} = 15$$

Value of C = 10

Hence answer is option B

Q10 Text Solution:

Total number of chocolates distributed among 'X', 'Y', and 'Z' = 100

Number of chocolates given to child 'X' = $(C - 5)\% \times 100 = C - 5$

Remaining chocolates the teacher left with = $100 - (C - 5) = 105 - C$

Number of chocolates given to child 'Y' = $(105 - C) \times (C + 5)\%$

Remaining chocolates the teacher left with = $(105 - C) - (105 - C) \times (C + 5)\% = (105 - C) \times (95 - C)\%$

Number of chocolates given to child 'Z' = $(105 - C) \times (95 - C)\% \times 25\%$

Remaining chocolates the teacher left with = $(105 - C) \times (95 - C)\% \times 75\%$

The teacher finally left with 54 chocolates.

$$\text{ATQ, } (105 - C) \times (95 - C)\% \times 75\% = 54$$

$$\Rightarrow (105 - C) \times (95 - C) = 7200$$

$$\Rightarrow 9975 - 200C + C^2 = 7200$$

$$\Rightarrow C^2 - 200C + 2775 = 0$$

$$\Rightarrow C^2 - 15C - 185C + 2775 = 0$$

$$\Rightarrow C \times (C - 15) - 185 \times (C - 15) = 0$$

$$\Rightarrow (C - 15) \times (C - 185) = 0$$

$$\Rightarrow C = 15 \text{ and } 185$$

C = 185 is not possible because the number of candies teacher left with will be negative.

Therefore, C = 15

Number of candies given to child 'Y' = $(105 - C) \times (C + 5)\% = 90 \times 20\% = 18$

If 'Y' gave 3 candies to his younger brother then the number of candies 'Y' left with = $18 - 5 = 13$

Hence the correct answer is 13

